

KCS

KCS stands for Knowledge-Centered Service, a methodology that aims to improve service delivery and knowledge management in service organizations. Some of the main points about KCS are:

- KCS is based on the principle that knowledge is a by-product of solving problems and that the knowledge should be captured, structured, and reused as part of the service process.
- KCS involves creating and maintaining knowledge articles that document the issues, solutions, and context of service interactions. These articles are stored in a knowledge base that can be accessed and updated by service agents and customers.
- KCS follows a set of practices and standards that guide the creation, improvement, and usage of knowledge articles. These include the KCS Loop, the KCS Article Quality Index, the KCS Solve and Evolve workflows, and the KCS Roles and Competencies.
- KCS benefits service organizations by reducing costs, increasing productivity, improving customer satisfaction, and enhancing organizational learning and innovation. KCS also enables service agents to develop their skills and expertise through knowledge sharing and collaboration.
- KCS is a registered trademark of the Consortium for Service Innovation, a non-profit organization that develops and promotes best practices for service and support. The Consortium provides resources, training, and certification for KCS practitioners and organizations.

Unit 1 - The cell concept, structure of prokaryotic, eukaryotic cells, plant cells and animal cells, Structure and function of cell membrane, cell organelles and their function, Basics of cell signaling, Structure and use of compound microscope, Basic chemical constituents of living body

- The cell concept is the idea that all living things are composed of one or more cells, and that cells are the basic units of structure and function in living organisms.
- Prokaryotic cells are cells that lack a nucleus and other membrane-bound organelles. They are usually unicellular and have a simple structure. Their DNA is found in the form of bundles called nucleoids .
- Eukaryotic cells are cells that have a nucleus and other membrane-bound organelles. They are usually multicellular and have a complex structure. Their DNA is found in the form of chromosomes inside the nucleus .
- Plant cells and animal cells are both examples of eukaryotic cells, but they have some differences. Plant cells have a cell wall, chloroplasts, and a large central vacuole, while animal cells do not. Animal cells have centrioles, lysosomes, and a small or absent vacuole, while plant cells do not.
- The cell membrane is a thin layer of phospholipids and proteins that sur-

rounds the cell and regulates the movement of substances in and out of the cell. It also maintains the cell's shape and communicates with other cells.

- Cell organelles are specialized structures within the cell that perform specific functions. Some examples of cell organelles are the nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, peroxisomes, cytoskeleton, cilia, and flagella.
- Cell signaling is the process by which cells communicate with each other and respond to external stimuli. Cells use chemical signals, such as hormones, neurotransmitters, and cytokines, to transmit information across the cell membrane or through the bloodstream. Cells also use physical signals, such as electrical impulses, light, and sound, to sense and react to their environment.
- A compound microscope is a device that uses two or more lenses to magnify small objects. It consists of an eyepiece, an objective lens, a stage, a light source, and a focusing mechanism. A compound microscope can be used to observe the structure and function of cells and other microscopic organisms.
- The basic chemical constituents of living body are the elements that make up the molecules of life. The most abundant elements in living organisms are carbon, hydrogen, oxygen, and nitrogen, which form the basis of organic compounds. Other elements, such as phosphorus, sulfur, calcium, iron, and magnesium, are also essential for various biological processes.

The cell concept

- The cell concept is the idea that **the cell is the fundamental unit of life** and that all living things are composed of one or more cells.
- The cell concept also implies that **cells arise from pre-existing cells** by a process of cell division, and that **cells carry genetic information** that is passed on to their offspring.
- The cell concept is one of the foundations of modern biology, and it helps us understand the structure, function, diversity and evolution of living organisms .

Structure of prokaryotic, eukaryotic, plant and animal cells

- Prokaryotic cells are cells that **lack a nucleus** and other membrane-bound organelles, such as bacteria and archaea .
- Prokaryotic cells have a **simple structure** that consists of a **cell membrane**, a **cell wall**, a **cytoplasm**, and **genetic material** (DNA or RNA) that is often circular and located in a region called the **nucleoid** .
- Prokaryotic cells may also have **external structures**, such as **flagella**, **pili**, or **capsules**, that help them move, attach, or protect themselves .
- Eukaryotic cells are cells that **have a nucleus** and other membrane-bound organelles, such as protists, fungi, plants and animals .

- Eukaryotic cells have a **complex structure** that consists of a **cell membrane**, a **cytoplasm**, and a **nucleus** that contains the **genetic material** (DNA) in the form of linear chromosomes .
- Eukaryotic cells also have various **organelles**, such as **mitochondria**, **endoplasmic reticulum**, **Golgi apparatus**, **lysosomes**, **peroxisomes**, **ribosomes**, **centrioles**, and **cytoskeleton**, that perform different functions, such as energy production, protein synthesis, transport, digestion, detoxification, and movement .
- Plant cells are a type of eukaryotic cells that **have some additional features** that distinguish them from animal cells .
- Plant cells have a **cell wall** that provides rigidity and support, a **large central vacuole** that stores water and other substances, and **plastids**, such as **chloroplasts**, that are involved in photosynthesis and other metabolic processes .
- Animal cells are a type of eukaryotic cells that **lack some features** that plant cells have, such as a cell wall, a central vacuole, and plastids .
- Animal cells have **smaller vacuoles** that are used for storage and transport, and **centrioles** that are involved in cell division and organization of the cytoskeleton .

Structure and function of cell membrane

- The cell membrane is a **thin, flexible, and selectively permeable** layer that surrounds the cytoplasm of a cell and separates it from the external environment .
- The cell membrane is composed of a **phospholipid bilayer** with embedded **proteins**, **carbohydrates**, and **cholesterol** molecules .
- The phospholipid bilayer consists of two layers of phospholipids, which are molecules that have a **hydrophilic** (water-loving) head and a **hydrophobic** (water-fearing) tail .
- The hydrophilic heads face the aqueous environments inside and outside the cell, while the hydrophobic tails face each other in the interior of the membrane .
- The proteins in the cell membrane can be **integral** or **peripheral** .
- Integral proteins span the entire membrane and are involved in **transport**, **signaling**, or **attachment** of the cell .
- Peripheral proteins are attached to the surface of the membrane

Structure of Prokaryotic Cells

- Prokaryotic cells are unicellular organisms that lack a nucleus and other membrane-bound organelles.
- Prokaryotic cells have a nucleoid region, which is an irregularly-shaped region that contains the circular, double-stranded DNA .
- Prokaryotic cells have a plasma membrane, which is an outer covering that separates the cell's interior from its surrounding environment.

- Prokaryotic cells have a cell wall, which is an outer layer that maintains the cell's shape, protects the cell interior, and prevents the cell from bursting when it takes up water . The cell wall of most bacteria contains peptidoglycan, a polymer of linked sugars and polypeptides.
- Some prokaryotic cells have a capsule, which is an additional outer covering that protects the cell when it is engulfed by other organisms, helps the cell adhere to surfaces, and prevents dehydration .
- Some prokaryotic cells have flagella, which are long, whip-like structures that help the cell move .
- Some prokaryotic cells have pili, which are short, hair-like structures that help the cell attach to other cells or surfaces, and exchange genetic material .
- Prokaryotic cells have ribosomes, which are small structures that synthesize proteins . Prokaryotic ribosomes are smaller and simpler than eukaryotic ribosomes.
- Some prokaryotic cells have plasmids, which are small, circular pieces of DNA that can replicate independently of the main chromosome and carry genes for special functions, such as antibiotic resistance .
- Some prokaryotic cells have inclusions, which are granules or vesicles that store nutrients, waste products, or other substances .
- Some prokaryotic cells have endospores, which are dormant, resistant structures that can survive harsh conditions and germinate into new cells when conditions improve .

Eukaryotic Cells

- Eukaryotic cells are one of the two types of cells that make up all living things, the other being prokaryotic cells.
- Eukaryotic cells have a nucleus that contains the genetic material (DNA) and is surrounded by a nuclear membrane.
- Eukaryotic cells also have other membrane-bound organelles that perform specific functions, such as mitochondria (energy production), endoplasmic reticulum (protein synthesis and transport), Golgi apparatus (modification and sorting of proteins), lysosomes (digestion and recycling of materials), and vacuoles (storage and regulation of water and solutes).
- Eukaryotic cells have a cytoskeleton that provides structure and support, and enables movement and transport within the cell. The cytoskeleton is composed of microtubules, microfilaments, and intermediate filaments.
- Eukaryotic cells are larger and more complex than prokaryotic cells, and can be divided into four major groups: protozoa, fungi, plants, and animals. These groups are classified into the domain Eukaryota.
- Eukaryotic cells differ from prokaryotic cells in several ways, such as having a nucleus, multiple chromosomes, membrane-bound organelles, and a larger size and volume.

Plant Cells

Plant cells are the basic units of life in green plants, photosynthetic eukaryotes of the kingdom Plantae. They have some distinctive features that differentiate them from animal cells and other types of eukaryotic cells. Here are some of the main characteristics of plant cells:

- Plant cells have a **cell wall** surrounding the cell membrane. The cell wall is made of cellulose, hemicelluloses, and pectin, and provides structural support, protection, and shape to the cell. The cell wall also allows the plant cell to withstand high internal pressure due to water uptake by osmosis. The cell wall has pores called plasmodesmata that allow the exchange of substances and signals between adjacent cells .
- Plant cells have **chloroplasts**, specialized organelles that contain the green pigment chlorophyll and are the site of photosynthesis. Photosynthesis is the process by which plants convert light energy into chemical energy, producing organic molecules such as glucose and oxygen. Chloroplasts have a double membrane and their own circular DNA, and they can divide independently of the cell .
- Plant cells have a large **central vacuole**, a water-filled sac enclosed by a membrane called the tonoplast. The central vacuole occupies most of the cell volume and serves various functions, such as storing nutrients, waste products, pigments, and toxins, maintaining the cell's turgor pressure, and regulating the cell's pH. The central vacuole can also contain enzymes that break down macromolecules or foreign substances .
- Plant cells have other common organelles that are also found in animal cells, such as the nucleus, the endoplasmic reticulum, the Golgi apparatus, the mitochondria, the ribosomes, the cytoskeleton, and the plasma membrane. These organelles perform essential roles in the cell's metabolism, gene expression, protein synthesis, energy production, and transport of molecules .
- Plant cells have some specialized cell-to-cell communication mechanisms that allow them to coordinate their activities and respond to environmental stimuli. For example, plant cells can use hormones, electrical signals, calcium waves, or reactive oxygen species to transmit information across the cell wall or the plasmodesmata. Plant cells can also sense and react to light, gravity, temperature, water, and pathogens through various receptors and signaling pathways .

: <https://www.britannica.com/science/plant-cell> https://en.wikipedia.org/wiki/Plant_cell
<https://biologydictionary.net/plant-cell/>

Animal Cells

- Animal cells are the basic structural and functional units of animal tissues and organs.
- Animal cells are eukaryotic cells, which means they have a nucleus and

membrane-bound organelles.

- Animal cells are surrounded by a plasma membrane, which forms a selective barrier that allows nutrients to enter and waste products to leave.
- Animal cells have various organelles that perform different functions, such as:
 - Nucleus: the control center of the cell that contains the genetic material (DNA) and regulates gene expression.
 - Cytoplasm: the fluid-filled space inside the cell that contains the organelles and various molecules.
 - Mitochondria: the powerhouses of the cell that produce energy (ATP) by cellular respiration.
 - Ribosomes: the sites of protein synthesis that are either free in the cytoplasm or attached to the endoplasmic reticulum.
 - Endoplasmic reticulum: a network of membranes that transports materials within the cell and synthesizes lipids and proteins.
 - Golgi apparatus: a stack of flattened membranes that modifies, sorts and packages proteins and lipids for secretion or delivery to other organelles.
 - Lysosomes: the digestive system of the cell that contains enzymes that break down macromolecules, worn-out organelles and foreign substances.
 - Centrosomes: the microtubule-organizing centers of the cell that play a role in cell division and movement.
 - Cytoskeleton: a network of protein fibers that provides shape, support and movement to the cell.
- Animal cells can be classified into different types based on their structure and function, such as:
 - Skin cells: the cells that form the outer layer of the body and protect it from external factors. They include melanocytes (pigment-producing cells), keratinocytes (keratin-producing cells), Merkel cells (touch receptor cells) and Langerhans cells (immune cells).
 - Muscle cells: the cells that contract and relax to produce movement. They can be skeletal (attached to bones), cardiac (found in the heart) or smooth (found in the walls of internal organs).
 - Blood cells: the cells that circulate in the blood and transport oxygen, carbon dioxide, nutrients, hormones and immune cells. They include red blood cells (erythrocytes), white blood cells (leukocytes) and platelets (thrombocytes).
 - Nerve cells: the cells that transmit electrical impulses and chemical signals to coordinate the activities of the body. They include neurons (nerve cells) and glial cells (support cells).
 - Fat cells: the cells that store energy in the form of lipids and regulate metabolism and hormone production. They include adipocytes (fat cells) and brown fat cells (heat-producing cells).

Structure and function of cell membrane

- The cell membrane, also known as the plasma membrane, is a thin semi-permeable membrane that surrounds the cytoplasm of a cell.
- The cell membrane has two main functions: first, to protect the integrity of the interior of the cell by allowing certain substances into the cell while keeping other substances out and, second, to be a gate allowing transport into the cell of essential nutrients and movement from the cell of waste products.
- The cell membrane also provides support and maintains the shape of the cell. It helps in regulating cell growth through the balance of endocytosis and exocytosis. It plays an important role in cell signalling and communication by using hormones, neurotransmitters, and other signaling molecules .
- The cell membrane is composed primarily of fatty-acid-based lipids and proteins, and carbohydrate groups that are attached to some of the lipids and proteins .
- The most abundant lipids in the cell membrane are phospholipids, which have a hydrophilic (water-loving) head and a hydrophobic (water-fearing) tail. Phospholipids form a bilayer in which the hydrophobic tails face each other and the hydrophilic heads face the aqueous environment on both sides of the membrane.
- Cholesterol is another lipid that is present in the cell membrane. It helps to maintain the fluidity and stability of the membrane by preventing the phospholipids from packing too closely or too loosely together.
- Proteins in the cell membrane have various functions, such as structural support, transport, enzymatic activity, signal transduction, cell recognition, and cell adhesion . Proteins can be classified as integral or peripheral, depending on their location and attachment to the membrane.
- Integral proteins are embedded in the phospholipid bilayer and span the entire thickness of the membrane. They can act as channels, carriers, receptors, or enzymes.
- Peripheral proteins are loosely attached to the surface of the membrane or to the integral proteins. They can act as anchors, enzymes, or regulators.
- Carbohydrate groups are attached to some of the lipids and proteins on the outer surface of the cell membrane. They form a layer called the glycocalyx, which helps to protect the cell from mechanical and chemical damage, and to mediate cell-cell recognition and communication.

Cell organelles and their function

- Cell organelles are small structures within the cytoplasm that carry out functions necessary to maintain homeostasis in the cell.
- Cell organelles are involved in many processes, such as energy production, protein synthesis, secretion, detoxification, and signal transduction .

- Cell organelles can be classified into two types: membranous and non-membranous.
- Membranous organelles are surrounded by a lipid bilayer membrane, similar to the plasma membrane that encloses the cell.
- Non-membranous organelles are not enclosed by a membrane and are directly in contact with the cytoplasm.
- Some of the major cell organelles and their functions are:
 - Plasma membrane: It is the outermost layer of the cell that separates it from the external environment. It is selectively permeable, meaning it allows some substances to pass through while blocking others. It also regulates the transport of materials into and out of the cell, and mediates cell-cell communication and recognition .
 - Cytoplasm: It is the jelly-like substance that fills the cell and contains the organelles. It provides a medium for the chemical reactions and molecular interactions that take place in the cell. It also maintains the shape and internal pressure of the cell .
 - Nucleus: It is the largest and most prominent organelle in eukaryotic cells. It contains the genetic material (DNA) of the cell, which is organized into chromosomes. It controls the expression of genes and the synthesis of proteins. It also regulates the cell cycle and cell division .
 - Nucleolus: It is a dense region within the nucleus that produces ribosomal RNA (rRNA), which is a component of ribosomes. Ribosomes are the sites of protein synthesis in the cell .
 - Endoplasmic reticulum (ER): It is a network of membranous tubules and sacs that extends from the nuclear envelope to the plasma membrane. It has two types: rough ER and smooth ER. Rough ER has ribosomes attached to its surface, and is involved in protein synthesis and modification. Smooth ER lacks ribosomes, and is involved in lipid synthesis, detoxification, and calcium storage .
 - Golgi apparatus: It is a stack of flattened membranous sacs that receives, sorts, modifies, and packages proteins and lipids from the ER. It then sends them to their final destinations, either within the cell or outside the cell via vesicles .
 - Lysosomes: They are spherical membranous sacs that contain digestive enzymes. They break down various substances, such as food particles, bacteria, viruses, worn-out organelles, and cellular debris. They also recycle the components of the degraded materials for reuse by the cell .
 - Mitochondria: They are oval-shaped membranous organelles that are the sites of cellular respiration. They convert the chemical energy stored in glucose and other nutrients into adenosine triphosphate (ATP), which is the main source of energy for the cell. They have their own DNA and ribosomes, and can replicate independently of

the nucleus .

- Chloroplasts: They are green-colored membranous organelles that are found only in plant cells and some algae. They are the sites of photosynthesis, which is the process of converting light energy into chemical energy. They also have their own DNA and ribosomes, and can replicate independently of the nucleus .
- Cytoskeleton: It is a network of protein filaments and tubules that provides structural support and shape to the cell. It also enables the movement of the cell and its organelles, and facilitates cell division and cell signaling .
- Centrioles: They are cylindrical structures composed of microtubules that are found in animal cells and some algae. They are located near the nucleus and form the poles of the spindle apparatus during cell division. They also organize the formation of cilia and flagella, which are hair-like or whip-like extensions of the plasma membrane that enable the movement of the cell or the surrounding fluid

Basics of cell signaling

- Cell signaling is the process of cellular communication within the body driven by cells releasing and receiving hormones and other signaling molecules.
- Cell signaling enables coordination within multicellular organisms and allows cells to respond to external stimuli .
- Cell signaling involves three stages: reception, transduction, and response .
 - Reception: A cell detects a signaling molecule from the outside of the cell. The signaling molecule binds to a specific receptor protein on the cell surface or inside the cell .
 - Transduction: When the signaling molecule binds the receptor, it changes the receptor protein in some way. This triggers a series of events or a signal relay cascade inside the cell, involving other molecules that relay and amplify the signal .
 - Response: Finally, the signal triggers a specific cellular response, such as a change in gene expression, enzyme activity, cell division, or cell death .
- Cell signaling can be classified into different forms based on the distance and type of signaling molecules involved:
 - Paracrine signaling: Cells secrete signaling molecules that affect nearby cells. For example, growth factors that stimulate cell division in neighboring cells.
 - Endocrine signaling: Cells secrete signaling molecules (hormones) that travel through the bloodstream and reach distant target cells. For example, insulin that regulates blood sugar levels in various tissues.
 - Autocrine signaling: Cells secrete signaling molecules that bind to

their own receptors and affect their own behavior. For example, cancer cells that stimulate their own growth and survival.

- Juxtacrine signaling: Cells directly contact each other and exchange signals through cell-cell junctions or membrane-bound molecules. For example, immune cells that recognize and eliminate foreign cells.
- Synaptic signaling: Neurons secrete signaling molecules (neurotransmitters) that cross the synaptic gap and bind to receptors on other neurons or target cells. For example, acetylcholine that stimulates muscle contraction.

Structure and use of compound microscope

- A compound microscope is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger.
- A compound microscope is mainly used for studying the structural details of cell, tissue, or sections of organs.
- The parts of a compound microscope can be classified into two: non-optical parts and optical parts.
- Non-optical parts include:
 - Base: The base is also known as the foot which is either U or horseshoe-shaped. It supports the microscope and houses the illuminator .
 - Arm: The arm is the curved part that connects the base and the head. It is used to carry the microscope .
 - Stage: The stage is the platform where the specimen is placed for observation. It has a hole in the center to allow light to pass through. It also has clips to hold the slide in place .
 - Coarse and fine adjustment knobs: These knobs are used to adjust the focus of the image by moving the stage or the objective lens up and down. The coarse adjustment knob is used for initial focusing, while the fine adjustment knob is used for precise focusing .
 - Condenser: The condenser is a lens system that collects and concentrates the light from the illuminator and directs it to the specimen. It has a knob to adjust its height and a diaphragm to control the amount of light .
- Optical parts include:
 - Eyepiece: The eyepiece is the lens that the observer looks through. It magnifies the image formed by the objective lens. It has a fixed magnification, usually 10x .
 - Objective lens: The objective lens is the lens that is closest to the specimen. It collects the light that passes through the specimen and forms an enlarged image. It has different magnifications, usually 4x, 10x, 40x, and 100x .
 - Nosepiece: The nosepiece is the rotating part that holds the objective lenses. It allows the observer to switch between different magnifica-

- tions by rotating the objective lens into position .
- Body tube: The body tube is the cylindrical part that connects the eyepiece and the nosepiece. It maintains the correct distance between the lenses for proper magnification .
- The working principle of a compound microscope is based on the refraction of light by the lenses. The light from the illuminator passes through the condenser and the specimen, and then enters the objective lens. The objective lens forms a magnified, inverted, and real image of the specimen. This image is further magnified by the eyepiece, which forms a virtual image that can be seen by the observer.
- The total magnification of a compound microscope is the product of the magnification of the objective lens and the eyepiece. For example, if the objective lens has a magnification of 40x and the eyepiece has a magnification of 10x, the total magnification is $40x \times 10x = 400x$.
- The advantages of a compound microscope are:
 - It can provide high magnification and resolution, which allows the observation of fine details of cells and tissues.
 - It can use different types of illumination, such as bright field, dark field, phase contrast, and fluorescence, to enhance the contrast and visibility of the specimen.
 - It can be equipped with various accessories, such as cameras, computers, and software, to capture, store, and analyze the images.
- The disadvantages of a compound microscope are:
 - It can only observe thin and transparent specimens, which may require special preparation, such as staining, cutting, and mounting.
 - It can cause chromatic and spherical aberrations, which are distortions of the image due to the imperfect refraction of light by the lenses.
 - It can have a limited depth of field, which is the range of distance within which the specimen is in focus. This means that only a small portion of the specimen can be seen clearly at a time.

Basic chemical constituents of living body

- The living body is composed of various chemical elements and compounds that are essential for its structure and function.
- The most abundant elements in the living body are oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus, which make up about 99% of the body mass .
- These elements are found in different types of organic molecules, such as carbohydrates, lipids, proteins, and nucleic acids, which are the building blocks of life .
- Carbohydrates are the main source of energy for the living body and are composed of carbon, hydrogen, and oxygen atoms. They include simple sugars, such as glucose and fructose, and complex polysaccharides, such as starch and cellulose.

- Lipids are hydrophobic molecules that store energy and form the cell membranes. They are composed of carbon, hydrogen, and oxygen atoms, and sometimes phosphorus and nitrogen atoms. They include fats, oils, waxes, steroids, and phospholipids.
- Proteins are the most diverse and functional molecules in the living body. They are composed of carbon, hydrogen, oxygen, nitrogen, and sulfur atoms, and sometimes other elements, such as iron and zinc. They are made of amino acids, which are linked by peptide bonds. They perform various roles, such as catalyzing reactions, transporting substances, providing structure, and regulating processes.
- Nucleic acids are the genetic material of the living body. They are composed of carbon, hydrogen, oxygen, nitrogen, and phosphorus atoms. They are made of nucleotides, which are linked by phosphodiester bonds. They store and transmit information, such as DNA and RNA.
- Water is the most abundant compound in the living body. It makes up about 60% of the body mass and is found in both the intracellular and extracellular fluids. It is composed of hydrogen and oxygen atoms, and has unique properties, such as high heat capacity, high surface tension, and polarity. It is essential for maintaining homeostasis, dissolving substances, facilitating reactions, and regulating temperature.
- Other elements and compounds that are found in trace amounts in the living body are called micronutrients. They include potassium, sodium, chlorine, magnesium, iron, iodine, copper, zinc, selenium, and fluorine. They are involved in various functions, such as maintaining fluid balance, transmitting nerve impulses, carrying oxygen, forming enzymes, and preventing diseases .

Unit 2 - Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

- Tissues are groups of similar cells that work together to perform a specific function.
- There are four main types of animal tissues: connective, nervous, muscle, and epithelial tissues.
- Connective tissue supports and binds other tissues, such as blood, cartilage, bone, and fat .
- Nervous tissue allows rapid communication between different parts of the body, such as the brain, spinal cord, and nerves.
- Muscle tissue can contract to allow movement of body parts, such as the heart, skeletal muscles, and smooth muscles .
- Epithelial tissue covers the surface of the body and the lining of organs, such as the skin, lungs, and digestive tract.
- There are three main tissue systems in plants: the epidermis, ground tissue, and vascular tissue.

- The epidermis is the outermost layer of cells that protects the plant from water loss, infection, and damage.
- Ground tissue is the bulk of the plant body that performs various functions, such as photosynthesis, storage, and support.
- Vascular tissue transports water and minerals from the roots to the leaves, and organic nutrients from the leaves to the rest of the plant, such as the xylem and phloem .
- Morphology is the study of the external form and structure of plants, such as the shape, size, color, and arrangement of plant parts.
- Anatomy is the study of the internal structure and organization of plants, such as the tissues, cells, and organelles.
- The root is the part of the plant that anchors it to the soil, absorbs water and minerals, and stores food.
- The stem is the part of the plant that supports the leaves, flowers, and fruits, and transports water and nutrients between them.
- The leaf is the part of the plant that is the main site of photosynthesis, where light energy is converted into chemical energy.
- The inflorescence is the part of the plant that bears the flowers, which are the reproductive organs of the plant.
- The flower is the part of the plant that contains the male and female reproductive structures, such as the stamen and the pistil.
- The fruit is the part of the plant that develops from the ovary of the flower after fertilization, and contains the seeds.
- The seed is the part of the plant that contains the embryo, which is the potential new plant, and the endosperm, which is the food supply for the embryo.

: Animal and plant tissues - Producing new cells - BBC Bitesize

Tissues - Structure and Function of Plant and Animal -Turito - US Learn
 Tissue Definition and Examples in Biology - ThoughtCo
 Tissues, organs, & organ systems (article) | Khan Academy
 Plant Morphology and Anatomy - an overview | ScienceDirect Topics
 Plant Parts and Functions | First and Second Grade Science Lesson For Kids - YouTube

Tissues in animal and plants

- Tissues are groups of similar cells that work together to perform a specific function.
- There are four main types of animal tissues: epithelial, connective, muscle, and nervous.
- There are three main types of plant tissues: epidermis, ground tissue, and vascular tissue.
- Some examples of animal tissues are blood, cartilage, muscle, and nerve.
- Some examples of plant tissues are xylem, phloem, parenchyma, and sclerenchyma.

- Animal tissues are mostly made of living cells, while plant tissues have many dead cells.
- Animal tissues do not have cell walls, while plant tissues have cell walls made of cellulose.
- Animal tissues are specialized for functions such as movement, communication, and protection, while plant tissues are specialized for functions such as transport, support, and photosynthesis .

Morphology

- Morphology is the study of the physical form and external structure of plants.
- Morphology includes both the vegetative and reproductive structures of plants.
- Morphology is useful in the visual identification of plants using dichotomous keys, herbarium samples, and digital databases.
- Morphology also helps to understand the development, variation, and evolution of plant form and function.

Root

- Root is the part of the plant that anchors it to the soil and absorbs water and minerals from the soil.
- Root can be classified into two types: taproot and fibrous root.
- Taproot is the main root that grows vertically downwards and gives rise to lateral branches called secondary roots.
- Fibrous root is a network of thin and branched roots that spread horizontally in the soil.
- Root has three regions: root cap, root meristem, and root hair.
- Root cap is a protective layer of cells that covers the tip of the root and secretes mucilage to lubricate the soil.
- Root meristem is a zone of actively dividing cells that produces new root cells and elongates the root.
- Root hair is a thin extension of the epidermal cell that increases the surface area for absorption of water and minerals.
- Root has three tissues: epidermis, cortex, and vascular cylinder.
- Epidermis is the outermost layer of cells that protects the root and absorbs water and minerals.
- Cortex is the middle layer of cells that stores food and transports water and minerals to the vascular cylinder.
- Vascular cylinder is the innermost layer of cells that contains the xylem and phloem.
- Xylem is the tissue that conducts water and minerals from the root to the stem and leaves.
- Phloem is the tissue that conducts food from the leaves to the root and other parts of the plant.

Stem

- Stem is the part of the plant that supports the leaves and flowers and transports water, minerals, and food.
- Stem can be classified into two types: herbaceous and woody.
- Herbaceous stem is soft and green and usually dies at the end of the growing season.
- Woody stem is hard and brown and persists for many years.
- Stem has three regions: node, internode, and bud.
- Node is the point on the stem where a leaf or a branch is attached.
- Internode is the part of the stem between two nodes.
- Bud is a small outgrowth on the stem that can develop into a leaf, a flower, or a branch.
- Stem has three tissues: epidermis, cortex, and vascular cylinder.
- Epidermis is the outermost layer of cells that protects the stem and prevents water loss.
- Cortex is the middle layer of cells that stores food and transports water and minerals to the vascular cylinder.
- Vascular cylinder is the innermost layer of cells that contains the xylem and phloem.
- Xylem and phloem are arranged in bundles called vascular bundles.
- In herbaceous stems, the vascular bundles are scattered throughout the stem.
- In woody stems, the vascular bundles are arranged in rings called annual rings.
- The annual rings indicate the age and growth rate of the stem.

Leaf

- Leaf is the part of the plant that performs photosynthesis and transpiration.
- Leaf can be classified into two types: simple and compound.
- Simple leaf is a single blade attached to the stem by a petiole.
- Compound leaf is a blade divided into leaflets attached to the stem by a rachis.
- Leaf has three regions: blade, petiole, and stipule.
- Blade is the flat and broad part of the leaf that contains the veins.
- Petiole is the stalk that connects the blade to the stem.
- Stipule is a small leaf-like structure at the base of the petiole.
- Leaf has three tissues: epidermis, mesophyll, and vascular tissue.
- Epidermis

Anatomy and Functions of Different Parts of Plants

- Plants are multicellular organisms that have two distinct organ systems: a shoot system and a root system.

- The shoot system consists of the stem, leaves, flowers, fruits and seeds, while the root system consists of the roots and root hairs.
- The stem is the main axis of the plant that supports the leaves and flowers. It also conducts water and minerals from the roots to the leaves, and food from the leaves to the rest of the plant.
- The stem can grow in length (primary growth) or in thickness (secondary growth). The stem has different types of tissues, such as epidermis, cortex, vascular tissue and pith.
- The leaf is the main organ of photosynthesis, the process of converting light energy into chemical energy. The leaf has a flat and thin structure that maximizes the surface area for light absorption.
- The leaf has an epidermis that protects the inner tissues and regulates gas exchange. The leaf also has a mesophyll that contains chlorophyll, the pigment that gives plants their green color and captures light energy. The leaf also has veins that transport water and food.
- The flower is the reproductive organ of flowering plants. It consists of four main parts: sepals, petals, stamens and carpels.
- The sepals are the outermost part of the flower that protect the bud. The petals are the colorful and attractive part of the flower that attract pollinators. The stamens are the male part of the flower that produce pollen, the male gamete. The carpels are the female part of the flower that contain the ovary, the ovule and the stigma.
- The fruit is the part of the flower that develops after fertilization. It protects the seeds and aids in their dispersal. The fruit can be fleshy or dry, depending on the type of plant.
- The seed is the part of the plant that contains the embryo, the endosperm and the seed coat. The embryo is the young plant that will grow into a new plant. The endosperm is the food reserve for the embryo. The seed coat is the protective layer that prevents water loss and infection.
- The root is the part of the plant that anchors the plant to the soil and absorbs water and minerals from the soil. It also stores food and hormones for the plant.
- The root can grow in length (primary growth) or in thickness (secondary growth). The root has different types of tissues, such as epidermis, cortex, endodermis, pericycle, vascular tissue and pith.
- The root hair is the extension of the root epidermis that increases the surface area for water and mineral absorption.

Root

- The root is the descending portion of the plant axis that is usually underground.
- The root system consists of the primary root and its branches, which may be modified for different functions.
- The root has four main functions: absorption of water and minerals from the soil, anchorage of the plant, storage of reserve food, and synthesis of

plant growth regulators.

- The root has a radial or concentric arrangement of tissues that determines the patterns and rates of nutrient transport from the soil to the vascular tissue.
- The root has three main regions: the root cap, the root meristem, and the root hair zone.
- The root cap is a protective layer of cells that covers the root tip and secretes mucilage to lubricate the soil.
- The root meristem is a region of actively dividing cells that produces new root cells and elongates the root.
- The root hair zone is a region of epidermal cells that have hair-like outgrowths that increase the surface area for absorption.
- The root has three main types of tissues: the epidermis, the cortex, and the stele.
- The epidermis is the outermost layer of cells that protects the root and absorbs water and minerals.
- The cortex is the middle layer of cells that stores food and transports water and minerals to the stele.
- The stele is the innermost layer of cells that consists of the vascular tissue (xylem and phloem) and the pericycle.
- The xylem is the tissue that conducts water and minerals from the root to the stem.
- The phloem is the tissue that conducts food from the stem to the root.
- The pericycle is the tissue that surrounds the vascular tissue and gives rise to lateral roots and secondary growth.
- The root has two main types of root systems: the taproot system and the fibrous root system.
- The taproot system is a type of root system that has a single, thick, main root (the taproot) that grows vertically downward and gives rise to smaller lateral roots.
- The fibrous root system is a type of root system that has many thin, branching roots that spread horizontally in the soil.
- The root may have different modifications for different purposes, such as storage, support, respiration, and parasitism.
- Some examples of modified roots are: tuberous roots, prop roots, pneumatophores, and haustoria.
- Tuberous roots are swollen roots that store food, such as sweet potato and carrot.
- Prop roots are adventitious roots that grow from the stem and provide support, such as in maize and banyan.
- Pneumatophores are aerial roots that grow from the submerged roots and facilitate gas exchange, such as in mangroves.
- Haustoria are parasitic roots that penetrate the host plant and absorb its nutrients, such as in mistletoe and dodder.

Stem

- The stem is an **axial organ** of the shoot system that supports and elevates the leaves, fruits, and flowers.
- The stem also arranges the leaves in a way that they receive direct sunlight for **photosynthesis**.
- The stem contains **vascular tissues** (xylem and phloem) that conduct water and nutrients across the plant.
- The stem also has **storage** and **synthetic** functions, as it can store water, starch, and other substances, and produce hormones and secondary metabolites.
- The stem has a **radial structure**, meaning that it has a central axis with tissues arranged in concentric layers.
- The stem bears **nodes** and **internodes**. Nodes are the regions where leaves are attached, while internodes are the portions between two nodes.
- The stem has a **dermal tissue** called the **epidermis** that covers and protects the stem. The epidermis may have a waxy **cuticle** that limits water loss, and may also have specialized structures such as hairs, glands, and stomata .
- The stem has a **ground tissue** called the **cortex** that lies between the epidermis and the vascular tissue. The cortex is mainly composed of **parenchyma** cells that have various functions, such as photosynthesis, storage, and secretion.
- The stem has a **vascular tissue** that consists of two types of conducting tissues: **xylem** and **phloem**. Xylem transports water and minerals from the roots to the shoots, while phloem transports organic molecules from the sources (such as leaves) to the sinks (such as fruits).
- The xylem and phloem are arranged in **vascular bundles** that are distributed in a ring-like pattern in the stem. The vascular bundles are surrounded by a layer of cells called the **bundle sheath**.
- The stem has a **meristematic tissue** called the **vascular cambium** that lies between the xylem and the phloem in each vascular bundle. The vascular cambium is a layer of undifferentiated cells that can divide and produce new xylem and phloem cells, resulting in the **secondary growth** of the stem.
- The stem also has a **meristematic tissue** called the **cork cambium** that lies under the epidermis in the cortex. The cork cambium produces **cork cells** that replace the epidermis and form a protective layer called the **periderm**.
- The stem can be classified into different types based on the shape, size, branching, and modification. Some examples of stem types are **herbaceous** (soft and green), **woody** (hard and lignified), **erect** (vertical), **prostrate** (horizontal), **climbing** (supported by other plants), **tuberous** (swollen and modified for storage), **bulbous** (short and fleshy), **thorny** (with sharp outgrowths), and **succulent** (with thick and fleshy tissues).

Leaf

- A leaf is a thin, flat organ of the plant that is mainly responsible for photosynthesis and transpiration.
- A leaf develops laterally at the node of the stem and originates from the shoot apical meristem.
- A leaf has three main parts: the blade, the petiole, and the stipules.
- The blade is the broad, flat part of the leaf that contains the veins and the mesophyll tissue.
- The petiole is the stalk that connects the blade to the stem.
- The stipules are small, leaf-like structures that are present at the base of the petiole in some plants.
- The morphology of the leaf can vary depending on the plant species, the environment, and the function.
- Some common types of leaf morphology are simple, compound, pinnate, palmate, lobed, serrated, and needle-like.
- Some leaf morphologies, such as folded leaves, the presence of pubescence, trichomes, and wax layers, play an explicit role in the removal of pollutants and favor the accumulation of particulate matter.
- The main function of a leaf is to produce food for the plant by photosynthesis.
- Photosynthesis is the process of converting light energy into chemical energy that can be used by the plant.
- Chlorophyll, the substance that gives plants their characteristic green color, absorbs light energy and transfers it to the reaction centers in the chloroplasts.
- The chloroplasts are the organelles that contain the thylakoid membranes, where the light-dependent reactions of photosynthesis take place.
- The light-dependent reactions produce ATP and NADPH, which are used in the light-independent reactions (also known as the Calvin cycle) in the stroma of the chloroplasts.
- The light-independent reactions use CO₂ and water to synthesize glucose and other organic molecules that can be stored or used by the plant.
- The internal structure of the leaf is protected by the leaf epidermis, which is continuous with the stem epidermis.
- The leaf epidermis consists of a single layer of cells that are covered by a cuticle, a waxy layer that prevents water loss and protects the leaf from pathogens and herbivores.
- The leaf epidermis also contains stomata, which are pores that allow gas exchange and transpiration.
- Transpiration is the process of water loss from the leaf surface due to evaporation.
- Transpiration helps to cool the leaf, maintain the water potential gradient, and transport minerals and nutrients from the roots to the shoots.
- The stomata are surrounded by guard cells, which regulate the opening and closing of the stomata in response to environmental cues, such as light,

temperature, humidity, and CO₂ concentration.

- The mesophyll tissue is the inner tissue of the leaf that contains the chloroplasts and is the site of photosynthesis.
- The mesophyll tissue is divided into two layers: the palisade layer and the spongy layer.
- The palisade layer is the upper layer of the mesophyll that consists of elongated cells that are tightly packed and contain many chloroplasts.
- The palisade layer is the main site of light absorption and photosynthesis.
- The spongy layer is the lower layer of the mesophyll that consists of irregularly shaped cells that are loosely packed and contain fewer chloroplasts.
- The spongy layer is the main site of gas exchange and transpiration.
- The spongy layer also contains air spaces that allow the diffusion of gases between the stomata and the mesophyll cells.
- The veins are the vascular bundles that transport water, minerals, and organic molecules between the roots and the leaves.
- The veins consist of two types of vessels: xylem and phloem.
- The xylem is the vessel that transports water and minerals from the roots to the leaves.
- The xylem is composed of dead, lignified cells that form hollow tubes that are reinforced by pits and rings.
- The phloem is the vessel that transports organic molecules, such as sugars and amino acids, from the leaves

Inflorescence

- An inflorescence is the arrangement of a cluster of flowers on a floral axis or peduncle .
- The inflorescence is of two main types, based on the mode of development and arrangement of flowers on the axis :
 - Racemose inflorescence (Indeterminate Inflorescence): The main axis continues to grow and produces flowers laterally in an acropetal succession (from base to apex). The older flowers are at the base and the younger ones are at the apex .
 - Cymose inflorescence (Determinate Inflorescence): The main axis terminates in a flower and produces lateral branches that also end in flowers. The flowers are produced in a basipetal succession (from apex to base). The older flowers are at the apex and the younger ones are at the base .
- There are also special types of inflorescence, such as:
 - Hypanthodium: A fleshy receptacle with a basal opening, enclosing numerous small unisexual flowers . Example: Ficus (Fig).
 - Cyathium: A cup-shaped involucre enclosing a single female flower and several male flowers . Example: Euphorbia.
 - Verticillaster: A pair of cymes arranged in opposite directions at the nodes of a square stem . Example: Salvia, Ocimum, Coleus.
 - Spadix: A fleshy spike of flowers enclosed by a large, often brightly

- colored, bract called spathe . Example: Colocasia, Alocasia, Anthurium.
- Capitulum or Head: A condensed or shortened raceme or spike of sessile flowers on a flattened receptacle . Example: Helianthus, Tridax, Calendula.
- There are also mixed types of inflorescence, such as:
 - Thyrsus: A mixed inflorescence with a main racemose axis and lateral cymose branches . Example: Jasmine, Lantana.
 - Anthocarpous: A mixed inflorescence with a main cymose axis and lateral racemose branches . Example: Nyctanthes, Clerodendron.

Flower

- A flower is a modified shoot that consists of a floral axis and four types of floral appendages: sepals, petals, stamens and carpels .
- The floral axis is the stem that bears the floral organs. It may be shortened or elongated, depending on the type of flower.
- The sepals are the outermost whorl of floral appendages. They are usually green and leaf-like, and protect the flower bud before it opens.
- The petals are the next whorl of floral appendages. They are usually colorful and showy, and attract pollinators to the flower.
- The stamens are the male reproductive organs of the flower. They consist of a filament and an anther, which produces pollen grains that contain the male gametes.
- The carpels are the female reproductive organs of the flower. They consist of an ovary, a style and a stigma. The ovary contains one or more ovules, which contain the female gametes. The style is a slender stalk that connects the ovary to the stigma. The stigma is the receptive surface that catches the pollen grains.
- The function of the flower is to facilitate sexual reproduction in flowering plants. The pollen grains from the anther of one flower are transferred to the stigma of another flower of the same species, either by wind, water, animals or self-pollination. This process is called pollination. The pollen grains then germinate on the stigma and grow a pollen tube that reaches the ovule in the ovary. The male gamete travels through the pollen tube and fuses with the female gamete in the ovule. This process is called fertilization. The fertilized ovule develops into a seed, and the ovary develops into a fruit. The fruit protects the seed and aids in its dispersal. The seed germinates and grows into a new plant, completing the life cycle of the flowering plant.

Fruit

- A fruit is the **soft, pulpy part of a flowering plant that contains seeds** .
- It is formed from the **ovaries of angiosperms** (plants that produce flow-

ers and seeds enclosed in a carpel) and is exclusive only to this group of plants .

- Fruits are a characteristic of flowering plants. As soon as **pollen and fertilization** occur, the ovary of a plant becomes a fruit and the ovules become a seed.
- Fruits are classified into four types namely: **simple, aggregate, multiple, and accessory fruits** .
- Simple fruits are derived from a single ovary of a single flower, such as apples, cherries, and tomatoes .
- Aggregate fruits are formed from several ovaries of a single flower, such as raspberries, blackberries, and strawberries .
- Multiple fruits are developed from the ovaries of several flowers that fuse together, such as pineapples, figs, and mulberries .
- Accessory fruits are fruits that include other parts of the flower besides the ovary, such as pears, peaches, and oranges .
- Fruits have various functions in the life cycle of plants, such as **protecting the seeds, dispersing the seeds, attracting pollinators, and providing food for animals and humans** .

Unit 2 - Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

- Tissues are groups of cells that have a similar structure and function in a multicellular organism.
- Tissues are classified into two main types: animal tissues and plant tissues .
- Animal tissues are further divided into four main types: epithelial, connective, muscle and nervous tissues .
- Plant tissues are further divided into two main types: meristematic and permanent tissues.
- Meristematic tissues are composed of actively dividing cells that are responsible for the growth of the plant.
- Permanent tissues are composed of mature cells that have a specific function and shape.
- Permanent tissues are further classified into three main types: simple, complex and special tissues.
- Simple tissues are composed of one type of cells that perform a common function, such as parenchyma, collenchyma and sclerenchyma.
- Complex tissues are composed of more than one type of cells that work together to perform a specific function, such as xylem and phloem.

- Special tissues are composed of cells that have a specialized structure or function, such as secretory tissues, epidermis and cork.
- Morphology is the study of the external form and structure of plants.
- Anatomy is the study of the internal structure and organization of plants.
- The main parts of a plant are the root, stem, leaf, inflorescence, flower, fruit and seed.
- The root is the part of the plant that anchors it to the soil and absorbs water and minerals from the soil.
- The stem is the part of the plant that supports the leaves, flowers and fruits and conducts water and nutrients throughout the plant.
- The leaf is the part of the plant that is the main site of photosynthesis and transpiration.
- The inflorescence is the part of the plant that is a group or cluster of flowers arranged on a stem.
- The flower is the part of the plant that is the reproductive organ and consists of four main parts: sepals, petals, stamens and carpels.
- The fruit is the part of the plant that is the mature ovary that contains the seeds.
- The seed is the part of the plant that is the fertilized ovule that contains the embryo and the endosperm.
- The functions of different parts of plants vary depending on their structure and location.
- Some of the functions of different parts of plants are:
 - The root functions to anchor the plant, absorb water and minerals, store food and hormones, and produce secondary roots.
 - The stem functions to support the plant, transport water and nutrients, store food and water, and produce buds and branches.
 - The leaf functions to photosynthesize, transpire, exchange gases, and regulate temperature and water loss.
 - The inflorescence functions to attract pollinators, display flowers, and protect the flowers from environmental stress.
 - The flower functions to produce gametes, facilitate pollination, and form fruits and seeds.
 - The fruit functions to protect the seeds, disperse the seeds, and attract seed dispersers.
 - The seed functions to germinate, produce a new plant, and store food and hormones.

Unit 3 - Classification of living organisms

- Classification is the process of grouping organisms based on their similarities and differences.
- Classification helps in understanding the diversity of life, the evolutionary relationships among organisms and the ecological roles of different groups of organisms.
- The most widely accepted system of classification is the five kingdom classification, proposed by R.H. Whittaker in 1969.
- The five kingdoms are Monera, Protista, Fungi, Plantae and Animalia.
- Each kingdom is further divided into smaller groups called phyla (or divisions in plants), classes, orders, families, genera and species.
- The smallest and most basic unit of classification is the species, which consists of a group of organisms that can interbreed and produce fertile offspring.
- The scientific name of a species is given by the binomial system of nomenclature, which uses two words: the genus name and the specific epithet.
- The genus name is capitalized and the specific epithet is lowercase. Both words are italicized or underlined. For example, the scientific name of human is *Homo sapiens*.
- The binomial system of nomenclature was developed by Carolus Linnaeus, a Swedish botanist, in the 18th century.
- Linnaeus also devised a system of animal and plant classification based on morphological characters, such as the number and arrangement of floral parts in plants and the presence or absence of backbone in animals.
- Animal and plant classification are two branches of biological classification that deal with the classification of animals and plants, respectively.
- Animal classification is based on the level of organization, symmetry, body cavity, segmentation, appendages, nervous system, circulatory system, respiratory system, excretory system, reproductive system and embryonic development of animals.
- Plant classification is based on the presence or absence of vascular tissue, seeds, flowers and fruits in plants.

Classification of living organisms

- Classification of living organisms is the scientific process of arranging living things into different groups and subgroups based on their similarities and differences .
- Classification helps to identify, name, study and compare living organisms in a systematic and logical manner .
- Classification also reveals the evolutionary relationships and history of living organisms .
- The basic unit of classification is the species, which is a group of organisms that can interbreed and produce fertile offspring .
- The most widely used system of classification is the binomial system of

nomenclature, which was developed by Carl Linnaeus in the 18th century .

- The binomial system assigns a unique two-part name to each species, consisting of the genus name and the specific epithet .
- For example, the scientific name of human is *Homo sapiens*, where *Homo* is the genus name and *sapiens* is the specific epithet .
- The binomial names are written in Latin or Latinized words, and are italicized or underlined .
- The genus name is capitalized, while the specific epithet is not .
- The binomial names are also followed by the name of the author who first described the species .
- For example, *Homo sapiens* Linnaeus, 1758 .
- The binomial system is based on the hierarchical classification of living organisms into different levels or ranks .
- The main levels or ranks are: kingdom, phylum, class, order, family, genus and species .
- Each level or rank is further subdivided into smaller and more specialized groups .
- For example, the kingdom Animalia is divided into several phyla, such as Chordata, Arthropoda, Mollusca, etc .
- The phylum Chordata is divided into several classes, such as Mammalia, Aves, Reptilia, etc .
- The class Mammalia is divided into several orders, such as Primates, Carnivora, Rodentia, etc .
- The order Primates is divided into several families, such as Hominidae, Cercopithecidae, Lemuridae, etc .
- The family Hominidae is divided into several genera, such as *Homo*, *Pan*, *Gorilla*, etc .
- The genus *Homo* is divided into several species, such as *Homo sapiens*, *Homo erectus*, *Homo neanderthalensis*, etc .
- The classification of living organisms is based on various criteria, such as morphology, anatomy, physiology, biochemistry, genetics, molecular biology, ecology, behavior, etc .
- The classification of living organisms is not fixed or final, but is constantly revised and updated based on new discoveries and evidence .
- The most widely accepted system of classification of living organisms is the five kingdom system, which was proposed by Robert Whittaker in 1969 .
- The five kingdoms are: Animalia, Plantae, Fungi, Protista and Monera .
- The kingdom Animalia includes all multicellular animals that are heterotrophic, eukaryotic and lack cell walls .
- The kingdom Plantae includes all multicellular plants that are autotrophic, eukaryotic and have cell walls made of cellulose .
- The kingdom Fungi includes all multicellular and unicellular organisms that are saprotrophic or parasitic, eukaryotic and have cell walls made of chitin .

- The kingdom Protista includes all unicellular and colonial organisms that are eukaryotic and have diverse modes of nutrition and locomotion .
- The kingdom Monera includes all prokaryotic organisms that lack

Five Kingdom Classification

- The five kingdom classification is a system of grouping living organisms based on their characteristics such as cell structure, mode of nutrition, reproduction, etc.
- It was proposed by Robert H. Whittaker in 1969 and is one of the most common ways to classify living beings .
- The five kingdoms are Monera, Protista, Fungi, Plantae and Animalia.
- The major groups and principles of classification in each kingdom are as follows:

Kingdom Monera

- It includes all the prokaryotic organisms, which are unicellular and lack a true nucleus and membrane-bound organelles.
- They can be autotrophic or heterotrophic, aerobic or anaerobic, and may have different shapes such as cocci, bacilli, spirilla, etc.
- They reproduce mainly by binary fission, but some can also exchange genetic material by conjugation, transformation, transduction, etc.
- They are classified into two major groups: Archaeobacteria and Eubacteria.
- Archaeobacteria are the ancient bacteria that live in extreme environments such as hot springs, salt lakes, marshes, etc. They have unique cell wall and membrane composition and metabolic pathways.
- Eubacteria are the modern bacteria that are found in diverse habitats such as soil, water, air, plants, animals, etc. They have a peptidoglycan cell wall and a variety of metabolic pathways.

Kingdom Protista

- It includes all the eukaryotic organisms that are unicellular or colonial, and do not fit into the other kingdoms.
- They have a true nucleus and membrane-bound organelles, and can be autotrophic or heterotrophic, motile or non-motile, free-living or parasitic, etc.
- They reproduce mainly by asexual methods such as binary fission, multiple fission, budding, etc., but some can also undergo sexual reproduction by conjugation, syngamy, etc.
- They are classified into several groups based on their features such as:
 - Chrysophytes: They are the golden algae (desmids) and diatoms that have golden-brown pigments and silica cell walls. They are mostly photosynthetic and aquatic.

- Dinoflagellates: They are the marine algae that have two flagella and cellulose plates. They can be photosynthetic or heterotrophic, and some can produce bioluminescence and toxins.
- Euglenoids: They are the freshwater algae that have one or two flagella and a pellicle. They can be photosynthetic or heterotrophic, and some can switch between the two modes.
- Slime moulds: They are the fungus-like protists that have a plasmodial or cellular body. They feed on decaying organic matter and can form spores for reproduction.
- Protozoans: They are the animal-like protists that have different modes of locomotion and nutrition. They can be classified into four subgroups: Amoeboids, Flagellates, Ciliates and Sporozoans.

Kingdom Fungi

- It includes all the eukaryotic organisms that are multicellular (except yeasts) and have a filamentous body called mycelium, which is composed of hyphae.
- They are heterotrophic and obtain their nutrients by absorption from the substrates they grow on. They secrete enzymes to break down complex organic molecules into simpler ones.
- They reproduce by asexual or sexual methods, and produce spores that are dispersed by wind, water, animals, etc.
- They are classified into four major groups based on their spore-producing structures and life cycles:
 - Zygomycetes: They are the fungi that form zygospores, which are thick-walled resting spores that result from the fusion of two compatible hyphae. They include the bread moulds, such as *Rhizopus*.
 - Ascomycetes: They are the fungi that form ascospores, which are haploid spores that are produced inside a sac-like structure called ascus. They include the yeasts, such as *Saccharomyces*, and the morels and truffles.
 - Basidiomycetes: They are the fungi that form basidiospores, which are haploid spores that are produced on a club-shaped structure called basidium. They include the mushrooms,

Major Groups for the Notes of the Unit 3 - Classification of Living Organisms

- Classification of living organisms is the process of grouping them based on their similarities and differences in various aspects such as morphology, anatomy, physiology, genetics, etc.
- Classification helps to organize the diversity of life, to understand the evolutionary relationships among different groups of organisms, and to

provide a universal system of nomenclature for naming and identifying them.

- One of the most widely accepted systems of classification is the five kingdom classification, proposed by Robert H. Whittaker in 1969 .
- The five kingdoms are Monera, Protista, Fungi, Plantae, and Animalia. They are based on the following criteria:
 - Cell structure: whether the cells are prokaryotic (lacking a true nucleus and membrane-bound organelles) or eukaryotic (having a true nucleus and membrane-bound organelles).
 - Mode of nutrition: whether the organisms are autotrophic (capable of making their own food by photosynthesis or chemosynthesis) or heterotrophic (dependent on other sources of organic matter for food).
 - Thallus organization: whether the body of the organism is unicellular (consisting of a single cell), colonial (consisting of a loose aggregation of similar cells), filamentous (consisting of a chain of similar cells), or multicellular (consisting of many cells of different types and functions).
 - Phylogenetic relationships: whether the organisms share a common ancestry and evolutionary history with other groups of organisms.
 - Reproduction: whether the organisms reproduce asexually (by binary fission, budding, fragmentation, spore formation, etc.) or sexually (by fusion of gametes, conjugation, etc.).
- The following table summarizes the main characteristics and examples of each kingdom:

Kingdom	Cell structure	Mode of nutrition	Thallus organization	Phylogenetic relationships	Reproduction	Examples
Monera	Prokaryotic	Autotrophic or heterotrophic	Unicellular, colonial, or filamentous	Bacteria and archaea	Asexual or sexual	E. coli, cyanobacteria, methanogens, etc.
Protista	Eukaryotic	Autotrophic or heterotrophic	Unicellular, colonial, or multicellular	Algae, protozoa, and slime molds	Asexual or sexual	Amoeba, Paramecium, Euglena, diatoms, etc.

Kingdom	Cell structure	Mode of nutrition	Thallus organization	Phylogenetic relationships	Reproduction	Examples
Fungi	Eukaryotic	Heterotrophic	Unicellular, colonial, or multicellular	Saprophytic or parasitic fungi	Asexual or sexual	Yeast, mushroom, bread mold, ring-worm, etc.
Plantae	Eukaryotic	Autotrophic	Multicellular	Photosynthetic plants	Asexual or sexual	Moss, fern, pine, rose, mango, etc.
Animalia	Eukaryotic	Heterotrophic	Multicellular	Animals	Asexual or sexual	Sponge, jelly-fish, earth-worm, cock-roach, frog, lion, etc.

- The five kingdom classification is not the only system of classification, and it has some limitations and drawbacks. For example, it does not account for the molecular and genetic differences among the organisms, it does not reflect the true evolutionary relationships among some groups of organisms, and it does not include the viruses, which are not considered as living organisms.
- Therefore, some alternative systems of classification have been proposed, such as the six kingdom system, the three domain system, and the phylogenetic system. However, the five kingdom system is still widely used and taught in biology courses.
- Systematic and binomial system of nomenclature are two methods of naming and identifying the organisms based on their scientific names.
- Systematic is the branch of biology that deals with the classification and nomenclature of living organisms. It aims to provide a logical and consistent system of naming and grouping the organisms based on their natural relationships and characteristics.

- Binomial system of nomenclature is the method of giving a scientific name to each organism, consisting of two parts: the genus name and the species name. For example, the scientific name of human is *Homo sapiens*, where *Homo* is the genus name and *sapiens* is the species name.
- The binomial system of nomenclature was proposed by Carl Linnaeus in 1758, and it is governed

Principles of classification in each kingdom

- Classification is the process of grouping living organisms into categories based on their similarities and differences.
- Classification helps to organize the diversity of life and to understand the evolutionary relationships among different groups of organisms.
- The most widely used system of classification is the five-kingdom classification proposed by R.H. Whittaker in 1969.
- The five kingdoms are: Monera, Protista, Fungi, Plantae and Animalia.
- Each kingdom has its own principles of classification based on certain characteristics, such as mode of nutrition, cell structure, body organization, reproduction and phylogeny.

Monera

- Monera is the kingdom of prokaryotic organisms, which lack a true nucleus and membrane-bound organelles.
- Monera includes bacteria, cyanobacteria (blue-green algae) and archaeobacteria (extremophiles).
- Monera is classified into two major groups: Eubacteria and Archaeobacteria.
- Eubacteria are the common bacteria that inhabit various environments and have diverse metabolic pathways. They are further classified based on their shape, cell wall structure, staining properties, oxygen requirement, mode of nutrition and genetic characteristics.
- Archaeobacteria are the ancient bacteria that live in extreme conditions, such as high temperature, high salinity, high acidity or high methane concentration. They have unique cell wall composition, membrane lipids and ribosomal RNA sequences. They are further classified based on their habitat, such as methanogens, halophiles, thermophiles and acidophiles.

Protista

- Protista is the kingdom of eukaryotic organisms, which have a true nucleus and membrane-bound organelles, but are not multicellular or have a simple level of multicellularity.
- Protista includes protozoa (animal-like protists), algae (plant-like protists) and slime molds (fungus-like protists).

- Protista is classified into three major groups: Protozoa, Algae and Slime molds.
- Protozoa are the heterotrophic protists that move by flagella, cilia or pseudopodia. They are further classified based on their locomotion, such as flagellates, ciliates, amoeboids and sporozoans.
- Algae are the autotrophic protists that have chlorophyll and other pigments for photosynthesis. They are further classified based on their pigmentation, cell wall composition, storage products and life cycle, such as green algae, red algae, brown algae and diatoms.
- Slime molds are the saprophytic protists that feed on decaying organic matter. They have a complex life cycle that involves a plasmodial stage (a mass of multinucleate cytoplasm) and a sporangial stage (a structure that produces spores). They are further classified based on their plasmodial or cellular nature, such as plasmodial slime molds and cellular slime molds.

Fungi

- Fungi is the kingdom of eukaryotic organisms that are heterotrophic and feed by absorption of organic matter.
- Fungi include molds, yeasts, mushrooms and lichens.
- Fungi is classified into four major groups: Zygomycota, Ascomycota, Basidiomycota and Deuteromycota.
- Zygomycota are the fungi that produce thick-walled sexual spores called zygospores. They have a coenocytic (non-septate) hyphae (filaments) and asexual spores called sporangiospores. They include bread molds and dung fungi.
- Ascomycota are the fungi that produce sexual spores called ascospores inside a sac-like structure called ascus. They have a septate hyphae and asexual spores called conidia. They include yeasts, penicillium, morels and truffles.
- Basidiomycota are the fungi that produce sexual spores called basidiospores on a club-shaped structure called basidium. They have a septate hyphae and asexual spores called basidiospores. They include mushrooms, puffballs, rusts and smuts.
- Deuteromycota are the fungi that have no known sexual stage and are classified based on their asexual spores and hyphal structure. They include ringworm, athlete's foot and cheese molds.

Plantae

- Plantae is the kingdom of eukaryotic organisms that are autotrophic and have chlorophyll for photosynthesis.
- Plant

Systematic and binomial system of nomenclature

- Systematic is the branch of science that deals with unique properties of species and groups to recognise, describe, name and arrange the diverse organisms according to an organised plan .
- Binomial nomenclature is a formal system of naming species of living things by giving each a name composed of two parts, both of which use Latin grammatical forms, although they can be based on words from other languages.
- The first part of the name is the genus, which is a group of closely related species. The second part is the specific epithet, which is unique to each species within the genus. The two parts are written in italics, with the genus name capitalised and the specific epithet in lowercase. For example, the scientific name of modern humans is *Homo sapiens* .
- The binomial system of nomenclature was developed by Carl Linnaeus, a Swedish botanist and zoologist, who published a large work, *Systema Naturae* (The System of Nature), in which he attempted to identify every known plant and animal.
- The advantages of binomial nomenclature are that it avoids confusion, provides a universal language, reflects evolutionary relationships, and allows for easy identification of organisms.
- The rules of binomial nomenclature are governed by the International Code of Zoological Nomenclature (ICZN) for animals and the International Code of Nomenclature for algae, fungi, and plants (ICNafp) for plants. Some of the basic rules are:
 - The scientific name must be unique for each species.
 - The scientific name must be based on Latin or latinised words.
 - The scientific name must be written in italics or underlined.
 - The genus name must be capitalised and the specific epithet must be in lowercase.
 - The genus name can be abbreviated to its first letter, followed by a period, when writing the name of more than one species in the same genus. For example, *H. sapiens* and *H. erectus* are both species in the genus *Homo*.
 - The author and date of publication of the scientific name can be added in parentheses after the name to indicate its origin and validity. For example, *Homo sapiens* Linnaeus, 1758.

Concept of animal and plant classification

- Animal and plant classification is the process of grouping living organisms into categories based on their similarities and differences.
- Classification helps to organize the diversity of life and to understand the evolutionary relationships among different groups of organisms.
- The most widely used system of classification is the Linnaean taxonomy, which was developed by Carl Linnaeus in the 18th century.

- Linnaean taxonomy assigns each organism a binomial name, consisting of a genus and a species, and places it in a hierarchy of ranks, such as kingdom, phylum, class, order, family, genus, and species.
- The Linnaean taxonomy has been modified and expanded over time, and now includes additional ranks, such as domain, subkingdom, superphylum, subphylum, superclass, subclass, superorder, suborder, infraorder, superfamily, subfamily, tribe, and subtribe.
- The Linnaean taxonomy is based on morphological and anatomical features, such as body shape, size, structure, and function. However, these features may not always reflect the true evolutionary history of organisms, as some groups may have convergent or divergent evolution, or may have undergone hybridization or horizontal gene transfer.
- Therefore, modern classification also uses molecular and genetic data, such as DNA sequences, to infer the phylogeny, or the evolutionary history, of organisms. Phylogenetic methods, such as cladistics, use shared derived characters, or synapomorphies, to construct branching diagrams, or cladograms, that show the evolutionary relationships among different groups of organisms.
- One of the major outcomes of phylogenetic analysis is the recognition of three domains of life: Bacteria, Archaea, and Eukarya. These domains are based on the differences in the structure and function of the cellular components, such as the cell membrane, the cell wall, the ribosomes, and the genetic material.
- The domain Eukarya includes four kingdoms: Protista, Fungi, Plantae, and Animalia. These kingdoms are based on the differences in the mode of nutrition, the presence or absence of a cell wall, the type of cell division, and the degree of multicellularity and differentiation.
- The kingdom Protista is a diverse and polyphyletic group of mostly unicellular eukaryotes that do not fit into the other kingdoms. Protists can be autotrophic, heterotrophic, or mixotrophic, and can have various modes of locomotion, such as flagella, cilia, or pseudopodia. Protists include algae, protozoa, slime molds, and water molds.
- The kingdom Fungi is a monophyletic group of heterotrophic eukaryotes that obtain their nutrients by secreting digestive enzymes and absorbing the organic molecules. Fungi have cell walls made of chitin, and reproduce by spores. Fungi include mushrooms, yeasts, molds, and lichens.
- The kingdom Plantae is a monophyletic group of autotrophic eukaryotes that use photosynthesis to produce organic molecules. Plants have cell walls made of cellulose, and reproduce by alternation of generations, which involves a multicellular haploid gametophyte and a multicellular diploid sporophyte. Plants include bryophytes, ferns, gymnosperms, and angiosperms.
- The kingdom Animalia is a monophyletic group of heterotrophic eukaryotes that ingest their food and digest it internally. Animals lack cell walls, and reproduce by sexual reproduction, which involves a multicellular diploid zygote and a multicellular haploid gamete. Animals include

sponges, cnidarians, flatworms, roundworms, annelids, mollusks, arthropods, echinoderms, and chordates.

Unit 4 - Concepts of alleles and genes, Mendelian Experiments, Cell cycle (Elementary Idea), mitosis and meiosis, techniques to study mitosis and meiosis. Origin of life, evidences for biological evolution, Mechanism of evolution –Variation (Mutation and Recombination), Natural Selection with examples, Types of natural selection

- Concepts of alleles and genes
 - Genes are chunks of DNA that contribute to particular traits or functions by coding for proteins that influence physiology .
 - Alleles are different versions of a gene, which vary according to the nucleotide base present at a particular genome location .
 - An individual's combination of alleles is known as their genotype .
 - In sexually reproducing organisms, each parent gives an allele for each gene, giving the offspring two alleles per gene.
 - Alleles can be dominant or recessive, meaning that they can mask or be masked by the other allele in a pair.
 - The expression of alleles in the phenotype (the observable traits or functions) depends on the mode of inheritance, which can be complete dominance, incomplete dominance, codominance, multiple alleles, or sex-linked.
- Mendelian Experiments
 - Mendelian experiments were conducted by Gregor Mendel, a 19th century monk and scientist, who studied the inheritance of traits in pea plants.
 - Mendel crossed purebred plants with different traits, such as flower color, seed shape, and pod color, and observed the patterns of inheritance in the offspring.
 - Mendel discovered the principles of segregation, independent assortment, and dominance, which form the basis of classical genetics.
 - Segregation states that each parent has two alleles for each gene, and that these alleles separate during gamete formation, so that each gamete carries only one allele for each gene.
 - Independent assortment states that the alleles of different genes are sorted independently of each other during gamete formation, resulting in various combinations of alleles in the offspring.
 - Dominance states that some alleles are dominant over others, and that the dominant allele determines the phenotype of the organism, regardless of the other allele.
- Cell cycle (Elementary Idea)
 - The cell cycle is the series of events that take place in a cell, leading to its division and duplication.

- The cell cycle consists of four phases: G1, S, G2, and M.
- G1 is the first gap phase, where the cell grows and performs its normal functions.
- S is the synthesis phase, where the cell replicates its DNA and prepares for division.
- G2 is the second gap phase, where the cell checks for errors in the DNA and makes any necessary repairs.
- M is the mitotic phase, where the cell divides into two daughter cells, each with the same number and type of chromosomes as the parent cell.
- Mitosis and meiosis
 - Mitosis and meiosis are two types of cell division that occur in eukaryotic cells.
 - Mitosis is the process of dividing a somatic cell (a body cell) into two identical daughter cells, each with the same number and type of chromosomes as the parent cell.
 - Mitosis consists of four stages: prophase, metaphase, anaphase, and telophase.
 - Prophase is the stage where the chromosomes condense and become visible, the nuclear envelope breaks down, and the spindle fibers form.
 - Metaphase is the stage where the chromosomes align at the equator of the cell, and the spindle fibers attach to the centromeres of the chromosomes.
 - Anaphase is the stage where the sister chromatids separate and move to opposite poles of the cell, pulled by the spindle fibers.
 - Telophase is the stage where the chromosomes reach the poles and decondense, the nuclear envelope reforms, and the spindle fibers disappear.
 - Meiosis is the process of dividing a germ cell (a reproductive cell) into four haploid daughter cells, each with half the number of chromosomes as the parent cell.
 - Meiosis consists of two rounds of division: meiosis I and meiosis II.

Concepts of alleles and genes

- Genes are chunks of DNA that contribute to particular traits or functions by coding for proteins that influence physiology .
- Alleles are different versions of a gene, which vary according to the nucleotide base present at a particular genome location .
- An individual's combination of alleles is known as their genotype .
- Bacteria, because they have a single ring of DNA, have one allele per gene per organism. In sexually reproducing organisms, each parent gives an allele for each gene, giving the offspring two alleles per gene.
- Mendel's "heritable factors," which we now call genes, are actually regions of DNA found on chromosomes. He discovered that some alleles are dominant and some are recessive, and that the offspring inherit one allele from

each parent for each trait.

- The expression of alleles in the phenotype depends on the type of inheritance, such as complete dominance, incomplete dominance, codominance, multiple alleles, sex-linked traits, etc.

Mendelian Experiments

- Mendelian experiments are the studies of inheritance patterns in pea plants conducted by Gregor Mendel, a monk and a scientist, in the 19th century .
- Mendel chose pea plants because they have distinct and easily observable traits, such as flower color, seed shape, and plant height, and they can self-fertilize or cross-fertilize .
- Mendel selected seven pairs of contrasting traits and performed controlled crosses between pure-breeding plants with different traits, such as purple and white flowers .
- Mendel observed the phenotypes (visible traits) of the offspring in each generation and counted the number of plants with each trait. He found that some traits disappeared in the first generation (F1) but reappeared in the second generation (F2) in a predictable ratio .
- Mendel proposed that each trait is controlled by a pair of factors (now called genes) that are inherited from the parents, one from each parent. He also proposed that each factor can exist in two forms (now called alleles) that are either dominant or recessive .
- Mendel's first law of segregation states that the two alleles of a gene separate during the formation of gametes (sperm and egg cells), so that each gamete carries only one allele of each gene .
- Mendel's second law of independent assortment states that the alleles of different genes are inherited independently of each other, as long as they are located on different chromosomes .
- Mendel's third law of dominance states that when two different alleles of a gene are present in an organism, only the dominant allele determines the phenotype, while the recessive allele has no visible effect.
- Mendel's experiments laid the foundation for modern genetics and revealed the basic principles of how traits are inherited and expressed in living organisms .

Cell cycle

- The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells .
- The cell cycle can be thought of as the life cycle of a cell.
- The cell cycle consists of four main phases: G1, S, G2, and M .
- G1 phase: The first gap phase, where the cell grows in size and synthesizes proteins and organelles .
- S phase: The synthesis phase, where the cell replicates its DNA and du-

plicates its chromosomes .

- G2 phase: The second gap phase, where the cell prepares for mitosis by checking the accuracy of DNA replication and making any necessary repairs .
- M phase: The mitotic phase, where the cell divides its nucleus and cytoplasm into two daughter cells .
- The cell cycle is regulated by various checkpoints and molecules that ensure the proper timing and sequence of events .
- The cell cycle is essential for growth, development, repair, and reproduction of cells .

Mitosis and Meiosis

Mitosis and meiosis are two types of cell division that occur in eukaryotic organisms. They have different purposes and outcomes, and they involve different numbers and arrangements of chromosomes.

- Mitosis is the process by which most cells in the body divide, such as skin cells, muscle cells, and nerve cells. It involves a single round of cell division, and produces two identical, diploid daughter cells. Diploid means that each cell has two sets of chromosomes, one from each parent. Mitosis is used for growth, repair, and asexual reproduction.
- Meiosis is the process by which gametes are produced, such as sperm and egg cells. It involves two rounds of cell division, and produces four different, haploid daughter cells. Haploid means that each cell has one set of chromosomes, half the number of the diploid cells. Meiosis is used for sexual reproduction, and it introduces genetic variation among the gametes.

The main stages of mitosis and meiosis are:

- Interphase: This is the phase before cell division, where the cell grows, replicates its DNA, and prepares for division. Interphase is the same for both mitosis and meiosis, and it consists of three subphases: G1, S, and G2.
- Prophase: This is the first phase of cell division, where the chromosomes condense, the nuclear envelope breaks down, and the spindle fibers form. In prophase of mitosis, the chromosomes are in pairs of sister chromatids, each with two identical copies of DNA. In prophase I of meiosis, the chromosomes are also in pairs of sister chromatids, but they also form pairs of homologous chromosomes, each with similar but not identical DNA. The homologous chromosomes exchange segments of DNA in a process called crossing over, which creates genetic diversity.
- Metaphase: This is the second phase of cell division, where the chromosomes align at the center of the cell. In metaphase of mitosis, the chromosomes are arranged randomly along the metaphase plate, with one sister chromatid on each side. In metaphase I of meiosis, the homologous chro-

mosomes are arranged in pairs along the metaphase plate, with one pair on each side. This is called independent assortment, and it also creates genetic diversity.

- **Anaphase:** This is the third phase of cell division, where the chromosomes separate and move to opposite poles of the cell. In anaphase of mitosis, the sister chromatids are pulled apart by the spindle fibers, and each chromatid becomes a single chromosome. In anaphase I of meiosis, the homologous chromosomes are pulled apart by the spindle fibers, and each pair of sister chromatids becomes a single unit.
- **Telophase:** This is the fourth and final phase of cell division, where the chromosomes reach the poles of the cell, the nuclear envelope reforms, and the chromosomes decondense. In telophase of mitosis, the cell has two identical nuclei, each with the same number and type of chromosomes as the original cell. In telophase I of meiosis, the cell has two different nuclei, each with half the number and different combinations of chromosomes as the original cell.
- **Cytokinesis:** This is the process that follows cell division, where the cytoplasm and the cell membrane split to form two separate cells. Cytokinesis occurs only in telophase during mitosis, while it occurs in both telophase I and telophase II during meiosis. In cytokinesis of mitosis, the cell divides into two identical, diploid daughter cells. In cytokinesis of meiosis, the cell divides into two different, haploid daughter cells after the first round of division, and then into four different, haploid daughter cells after the second round of division.

The following diagrams illustrate the main stages of mitosis and meiosis:

Mitosis:[max_bytes\(150000\):strip_icc\(\)/mitosis-373389-final-5c46f0fbc9e77c0001a9c6f4.png](https://www.thoughtco.com/differences-between-mitosis-and-meiosis-373390)

Source: <https://www.thoughtco.com/differences-between-mitosis-and-meiosis-373390>

[Meiosis](<https://www.thoughtco.com/thmb/6wZ0w0w4s4s4s4s4s4>)

Techniques to study mitosis and meiosis

Mitosis and meiosis are two types of cell division that are essential for life. Mitosis produces two identical daughter cells from one parent cell, while meiosis produces four genetically different daughter cells from one parent cell. Both processes involve the duplication and separation of chromosomes, but they differ in the number of divisions, the alignment of chromosomes, and the resulting genetic variation.

To study mitosis and meiosis, you can use the following techniques:

- **Drawing:** Drawing both processes out on paper is a good practice exercise to deepen your understanding of mitosis and meiosis. Don't forget to label the phases, and write a brief description of what is occurring in each.

- **Simulation:** You can also use online simulations or animations to review the material and to quiz yourself. These tools can help you visualize the movements of chromosomes and the changes in the cell during each phase.
- **Squash preparation:** You can prepare a slide of onion root tips to observe the stages of mitosis under a microscope. Onion root tips are ideal because they have a high rate of cell division and the chromosomes are large and easy to see. To make a squash preparation, you need to fix the root tips in a solution of acetic acid and alcohol, stain them with a dye such as acetocarmine, and gently squash them between a slide and a cover slip.
- **Dissection:** You can prepare a slide of grasshopper testes to observe the stages of meiosis under a microscope. Grasshopper testes are suitable because they have a large number of cells undergoing meiosis and the chromosomes are distinct and visible. To make a dissection, you need to chloroform a grasshopper, remove its testes in a saline solution, fix them in a solution of acetic acid, chloroform, and alcohol, stain them with a dye such as acetocarmine, and mount them on a slide with a cover slip.

These techniques can help you study mitosis and meiosis in a practical and interactive way. You can also use online resources such as Khan Academy to learn more about the concepts and mechanisms of these processes.

Origin of life

- The origin of life is the scientific question of how life first emerged from non-living matter on Earth, or possibly other planets.
- There is no direct evidence or eyewitness account of the origin of life, so scientists rely on natural laws, experiments, and observations to propose plausible hypotheses and testable predictions.
- One of the main challenges of the origin of life is to explain how the organic molecules that define life, such as amino acids, nucleotides, sugars, and lipids, were formed and concentrated in the prebiotic environment.
- Another challenge is to explain how these organic molecules were assembled into macromolecules, such as proteins and nucleic acids, that can store and transmit genetic information and perform catalytic functions.
- A third challenge is to explain how these macromolecules were organized into self-replicating systems that could evolve and adapt to changing conditions.
- Some of the possible scenarios for the origin of life include:
 - The primordial soup hypothesis, which suggests that organic molecules were synthesized by energy sources such as lightning, ultraviolet radiation, or volcanic activity in a reducing atmosphere, and accumulated in a warm ocean or pond.
 - The hydrothermal vent hypothesis, which suggests that organic molecules were synthesized by geochemical reactions in the high-pressure and high-temperature environment of deep-sea vents, and

formed complex structures in the mineral-rich fluids and surfaces.

- The extraterrestrial hypothesis, which suggests that organic molecules or even primitive life forms were delivered to Earth by meteorites, comets, or interstellar dust, and seeded the planet with the building blocks of life.
 - The RNA world hypothesis, which suggests that RNA was the first macromolecule to emerge, and acted as both a genetic material and a catalyst for biochemical reactions, and later gave rise to DNA and proteins.
 - The lipid world hypothesis, which suggests that lipid-like molecules were the first macromolecules to emerge, and formed vesicles that could enclose other molecules and undergo self-replication and evolution.
- The origin of life is an active and interdisciplinary field of research that involves biology, chemistry, physics, geology, astronomy, and philosophy. It aims to uncover the natural processes that led to the emergence of life on Earth, and possibly elsewhere in the universe.

Evidences for biological evolution

Biological evolution is the change in the inherited characteristics of populations of organisms over time. There are many lines of evidence that support the theory of evolution, such as:

- **Fossils:** Fossils are the preserved remains or traces of organisms that lived in the past. Fossils can show how organisms have changed over time, or how they are related to each other. For example, fossils of dinosaurs, whales, and horses show how these animals have evolved from their ancestors .
- **Comparative anatomy:** Comparative anatomy is the study of the similarities and differences in the structures of different organisms. Comparative anatomy can reveal how organisms are related by descent from a common ancestor, or how they have adapted to different environments. For example, the forelimbs of humans, bats, whales, and cats have the same basic bones, but different shapes and functions. These are called homologous structures, and they indicate a common ancestry . On the other hand, the wings of birds, bats, and insects have different structures, but similar functions. These are called analogous structures, and they indicate convergent evolution .
- **Embryology:** Embryology is the study of the development of embryos, or the early stages of life. Embryology can show how organisms share common features or stages in their development, or how they differ from each other. For example, the embryos of vertebrates, such as fish, amphibians, reptiles, birds, and mammals, have similar features, such as a tail, gill slits, and a notochord. These features indicate a common ancestry . On the other hand, the embryos of different plants, such as mosses, ferns,

and flowering plants, have different features, such as spores, seeds, and flowers. These features indicate divergent evolution.

- **Molecular biology:** Molecular biology is the study of the molecules that make up the cells and genes of living organisms. Molecular biology can show how organisms share similar DNA sequences, proteins, or biochemical pathways, or how they differ from each other. For example, the DNA sequences of humans and chimpanzees are about 98% identical, indicating a close evolutionary relationship . On the other hand, the DNA sequences of humans and bacteria are very different, indicating a distant evolutionary relationship .
- **Biogeography:** Biogeography is the study of the distribution of organisms across the Earth. Biogeography can show how organisms have evolved in response to geographic isolation, climate change, or continental drift. For example, the marsupials, such as kangaroos, koalas, and wombats, are mostly found in Australia, indicating that they evolved from a common ancestor that was isolated from other continents . On the other hand, the cacti, such as saguaro, prickly pear, and barrel cactus, are found in different continents, such as North America, South America, and Africa, indicating that they evolved independently from different ancestors .
- **Direct observation:** Direct observation is the witnessing of evolutionary changes in living organisms over a short period of time. Direct observation can show how organisms have evolved in response to natural or artificial selection, genetic drift, or gene flow. For example, the bacteria, such as *E. coli* and *Staphylococcus*, have evolved resistance to antibiotics, indicating that they have adapted to a new environment . On the other hand, the finches, such as Darwin's finches, have evolved different beak shapes and sizes, indicating that they have diversified into different niches .

Mechanism of evolution

- Evolution is the process by which modern organisms have descended from ancient ancestors.
- Evolution is responsible for both the remarkable similarities and the amazing diversity of life on Earth.
- Evolution occurs when the frequency of alleles (variants of genes) in a population changes over generations.
- There are several mechanisms that can cause evolution, such as natural selection, mutation, genetic drift, gene flow, and non-random mating .
- Natural selection is the process by which individuals with certain traits have higher survival and reproduction rates than others in a given environment .
- Natural selection can result in adaptation, which is the increase of beneficial traits in a population over time .
- Natural selection can also lead to speciation, which is the formation of new species from existing ones .
- Mutation is the change in the DNA sequence of a gene or a chromosome .

- Mutation can introduce new alleles or traits into a population, or alter the existing ones .
- Mutation can be caused by errors in DNA replication, exposure to radiation, chemicals, or viruses, or other factors .
- Mutation can be beneficial, neutral, or harmful, depending on the effect on the organism's fitness .
- Genetic drift is the random change in allele frequency in a population due to chance events .
- Genetic drift can cause loss of genetic variation, fixation of alleles, or divergence of populations .
- Genetic drift is more pronounced in small populations, where random sampling effects are stronger .
- Gene flow is the movement of alleles or genes from one population to another due to migration, dispersal, or interbreeding .
- Gene flow can increase genetic variation within populations, or reduce genetic differences between populations .
- Gene flow can also introduce new alleles or traits into a population, or alter the existing ones .
- Non-random mating is the preference of individuals to mate with others who have certain traits or characteristics .
- Non-random mating can affect the distribution of alleles or genotypes in a population, but not the overall allele frequency .
- Non-random mating can be positive (assortative) or negative (disassortative), depending on whether similar or dissimilar individuals are preferred .
- Non-random mating can also be influenced by sexual selection, which is a form of natural selection that favors traits that increase mating success .
- Sexual selection can result in sexual dimorphism, which is the difference in appearance or behavior between males and females of the same species .

Variation

- Variation is the difference between individuals of the same species, caused by genetic and environmental factors .
- Genetic variation arises due to mutation and recombination during meiosis. Mutation is a change in the DNA sequence of a gene, which may result in a new allele or trait. Recombination is the exchange of genetic material between homologous chromosomes during meiosis, which may produce new combinations of alleles or traits.
- Environmental variation arises due to the effect of environmental factors on the expression of the genetic potentials. Environmental factors include temperature, light, nutrition, stress, etc. These factors may influence the phenotype or observable characteristics of an individual, such as height, weight, color, behavior, etc.
- Variation is important for biological evolution, as it provides the raw ma-

terial for natural selection. Natural selection is the process by which individuals with certain traits that are better adapted to their environment survive and reproduce more than others, leading to changes in the frequency of alleles or traits in a population over time.

- Types of natural selection include directional selection, stabilizing selection, and disruptive selection. Directional selection favors one extreme of a trait over the other, leading to a shift in the mean value of the trait in a population. Stabilizing selection favors the intermediate value of a trait over the extremes, leading to a reduction in the variation of the trait in a population. Disruptive selection favors both extremes of a trait over the intermediate, leading to a split in the population into two or more subgroups with different traits.
- Examples of natural selection include the evolution of antibiotic resistance in bacteria, the evolution of industrial melanism in peppered moths, the evolution of beak size and shape in Darwin's finches, etc.

Mutation and Recombination

- Mutation is a change in the DNA sequence of an organism that may or may not affect its phenotype.
- Recombination is the exchange of genetic material between different molecules of DNA, such as during crossing over in meiosis.
- Mutation and recombination are two sources of genetic variation that contribute to biological evolution.
- Mutation can occur spontaneously due to errors in DNA replication or repair, or induced by external factors such as radiation, chemicals, or viruses.
- Mutation can be classified into different types based on the size and location of the change, such as point mutation, insertion, deletion, inversion, duplication, or translocation.
- Mutation can also be classified into different types based on the effect on the protein product, such as silent, missense, nonsense, or frameshift mutation.
- Mutation can have different consequences for the organism, such as beneficial, neutral, deleterious, or lethal.
- Recombination can occur between homologous chromosomes during meiosis, or between non-homologous chromosomes or DNA molecules during transposition, viral infection, or genetic engineering.
- Recombination can generate new combinations of alleles and genes that increase the diversity of the gene pool.
- Recombination can also affect the linkage and segregation of genes, and the frequency and distribution of alleles in a population.